

Judy Hazem Abdelsattar

Giza, Egypt

☎ +20 1007471601 ✉ judyhazem2675@gmail.com 🔗 linkedin

Profile Summary

AI Engineer with a strong foundation in machine learning, natural language processing, and AI-powered systems. Experienced in building intelligent applications using Python, FastAPI, and modern AI frameworks. Skilled in developing end-to-end solutions, from data processing and model development to deployment. Passionate about leveraging AI to solve real-world problems and continuously learning new technologies.

Technical Skills

Languages: Python, Java, C, PHP, Dart, Solidity, JavaScript

Frameworks:

AI / Machine Learning:

Pandas, NumPy

TensorFlow, Keras, Scikit-learn

Hugging Face Transformers, Sentence Transformers, spaCy

Flair NLP, GLiNER

Statsmodels

Web / App Development:

React, Node.js, Flutter

Tools/Platforms: Git, GitHub, Microsoft Azure, PowerBI, DVC, MLflow

Concepts:

- Object-Oriented Programming
- Data Structures
- Machine Learning
- Deep Neural Networks
- Time Series Forecasting
- Computer Vision
- Natural Language Processing
- Speech Processing
- Data Wrangling & Visualization
- Big Data Analysis
- MLOps

Education

Bachelor of Artificial Intelligence & Data Science, Eslsca University

Specialization: Artificial Intelligence & Machine Learning

GPA: 3.88 | **ECTS:** 9.7

Professional Experience

FinTech & Digital Transformation Intern, Banque Misr

July 2025 – September 2025

- Gained exposure to fintech solutions and digital banking systems.
- Assisted in understanding electronic payment platforms and mobile banking services
- Observed digital transformation practices in banking operations.

Projects

GradVoice – Graduation Project

Python, FastAPI, Azure OpenAI, Whisper, NLP

- Designed and developed an end-to-end AI system for CV–Job Description analysis to support intelligent recruitment decision-making.
- Developed a cross-platform mobile application enabling users to upload CVs, receive evaluations, and interact with the system seamlessly.
- Built a semantic NLP pipeline to extract, structure, and compare candidate skills, experience, and qualifications

with job requirements.

- Integrated LLM-based reasoning using Azure OpenAI (GPT models) to generate ATS-style scoring, structured candidate assessments, and personalized feedback.
- Developed a real-time AI-guided interview system via browser microphone using Web Speech API for live candidate interaction.
- Implemented real-time speech-to-text transcription with speaker labeling and timestamped dialogue tracking.
- Built a voice-enabled AI assistant using Whisper (STT), Azure TTS, and Voice Activity Detection (VAD) for efficient real-time interaction.
- Enabled barge-in functionality, allowing users to interrupt AI responses mid-conversation for natural, human-like dialogue.
- Implemented semantic matching and ranking using Sentence Transformers and NER models (GLiNER, Flair NLP) to identify top candidate-job matches.
- Designed post-session analytics dashboards including full transcripts, AI insights, and structured interview summaries.
- Documented system architecture, APIs, and modular AI workflows to ensure scalability, maintainability, and extensibility.

Netflix Data Analysis & Business Insights Project

Python, Pandas, Power BI

- Conducted large-scale data analysis on Netflix datasets (8,800+ records, multi-source) to evaluate subscriber growth, revenue trends, and global market dynamics.
- Performed data preprocessing, including cleaning, deduplication, merging, and transformation using Power Query and Python.
- Designed and developed interactive Power BI dashboards to visualize key metrics such as revenue, subscriber trends, content distribution, and regional performance.
- Analyzed pre- and post-COVID trends to assess the impact on content production, user behavior, and platform growth.
- Explored audience segmentation and content preferences to identify dominant genres and engagement patterns.
- Evaluated regional pricing strategies and content availability to uncover mismatches between subscription cost and library value.
- Generated data-driven insights and recommendations for content strategy, pricing optimization, and regional expansion.

Healthcare Data Wrangling & Visualization Project

Python, SQL, Tableau

- Developed a comprehensive data wrangling pipeline on real-world healthcare data (10,000+ records) to prepare and structure data for analysis.
- Performed extensive data cleaning and preprocessing, including handling missing values, outliers, and inconsistent data types using Python (Pandas, NumPy).
- Applied data transformation techniques such as normalization, encoding, aggregation, and feature restructuring to improve data quality and usability.
- Designed and executed SQL queries to integrate and manipulate data from multiple sources, enabling efficient data retrieval and analysis.
- Conducted exploratory data analysis (EDA) to generate statistical insights and identify patterns in healthcare data.
- Built interactive Tableau dashboards to visualize key trends, correlations, and healthcare metrics, ensuring clear and intuitive data storytelling.
- Generated analytical insights and documented findings to support data-driven decision-making and highlight healthcare patterns.

Arabic Tweet Sentiment Classification (Kaggle Competition)

Python, NLP, Transformers

- Developed a multi-class sentiment classification model for Arabic tweets (NEG, POS, OBJ, NEUTRAL) using Natural Language Processing techniques.
- Preprocessed and cleaned Arabic text data, including normalization, tokenization, and handling linguistic variations.
- Trained and evaluated machine learning and transformer-based models for sentiment prediction.
- Optimized model performance using Macro F1-score to handle class imbalance and ensure balanced evaluation across all categories.
- Applied feature engineering and model tuning techniques to improve classification accuracy.
- Conducted comparative analysis of different models to select the best-performing approach.

Garbage Image Classification using Deep Learning

Python, TensorFlow, CNN

- Developed and compared multiple deep learning models (DNN, custom CNN, and pretrained CNN) for multi-class

image classification (10 categories).

- Built a custom CNN architecture to extract spatial features and improve classification performance.
- Applied transfer learning using InceptionResNetV2 with fine-tuning to enhance model accuracy.
- Implemented data preprocessing and augmentation techniques to improve generalization on a limited dataset (3,000 images).
- Achieved 92.5% accuracy with the pretrained model, significantly outperforming baseline models.
- Evaluated model performance using precision, recall, F1-score, and confusion matrix analysis.

Certificates

- Best Data Analyst Award (School of Computing & Digital Technology, ESLSCA University)

Languages & Soft Skills

Languages: Arabic (Native), English (Fluent)

Soft Skills: Time Management, Communication, Presentation, Problem Solving, Teamwork, Decision Making, Continuous Learning, Adaptability, Resilience